

CS211: Computer Organisation

Semester I, 2025

Assignment #1 (Weight: 5%)

Due: 6:00pm, Friday 4th April, 2025

Learning Outcome: Solve computational problems using assembly and machine language.

Wombat 1.1

A function is defined as follows:

$$f(n) = \begin{cases} n^2 & \text{for } n < 1 \\ n - 5 & \text{for } n = 1 \\ n^2 + 3n + 1 & \text{for } n > 1 \end{cases}$$

Write **Wombat1 machine language program** that reads an input integer value n , and computes and displays the output of the above function $f(n)$.

In the binary version of your programs that can be executed on the CPUSim simulator, you are also required to write as comments the line number/address, equivalent decimal version of the binary and appropriate comments to explain each line in your program. The decimal version of the program represents each binary line as two decimal values (first value is the opcode – 4 bits; second value is the operand/memory address – 12 bits). Therefore, each line of your program should be formatted as follows:

Binary-number, Semi-colon, Line-number/address, Decimal-opcode, Decimal-operand/address, Comments

As an example, suppose you are writing a program to read and display a number. This program could be written as follows:

```

0011000000000000 ; 0:  3 0  read n into accumulator
00100000000001010 ; 2:  2 10 store n at location 10
00010000000001010 ; 4:  1 10 load n from location 10
01000000000000000 ; 6:  4 0  output n
00000000000000000 ; 8:  0 0  stop
00000000000000000 ; 10: 0 0  storage location for n

```

Submission Guidelines

- **This assignment is to be completed individually.**
- Submit only one file which has the wombat program.
- Only **one** submission is required – rename the file as your id number.
- A **softcopy** of your assignment should be submitted through **Moodle Drop Box** using. Submitted assignment should undergo plagiarism check using turn-it-in. The maximum similarity check allowed is 20%.
- **Incorrect/incomplete submissions or any assignments caught as plagiarized work will receive a mark of zero (0).**
- Late submissions will **not** be accepted unless prior arrangements have been made with the lecturer and supplemented by documentary evidence.
- Do not submit any piece of assignment work on Moodle Discussion Forums.

The following rubric will be used to mark your assignment. This rubric assumes **no late submission**.

CBOK	Unsatisfactory (0%-49%)	Satisfactory (50% - 75%)	Good (76% - 100%)
Programming	I. Plagiarism II. Poor indentation, hard to read and follow the code III. Lots of bugs and/or errors IV. Program produces unexpected output VII. Hard coding of data in the program. VIII. Poor structure	I. Able to write a simple code for a well-defined problem II. Use of basic standard programming practices such as commenting, indentation etc. III. computer program produces correct output	I. All satisfactory and demonstrate very good programming skills.

	Unsatisfactory (1)	Satisfactory (2)	Good (3)	Excellent (4)	Enter 1,2,3,4	Weight	Total Score
Program Design	Lengthy program with lots of unused storages, unnecessary instructions. Difficult and inefficient program. Rubric: I - VIII	Lengthy program with some unused storages, unnecessary instructions. Program is logical but not an efficient one. Rubric: I - III	Program has some unnecessary instructions. program is logical and easy to follow. Rubric: I	Program has no unnecessary instructions and unused storages. program is efficient, easy to understand. Rubric: I		15%	
Comments	some comments included.	Some comments with some description of machine instruction.	Clearly commented with specific purpose of each machine instruction.	Clear and effective comments with description and purpose of each instruction.		5%	
Runtime	Program runs but does not produce any output Rubric: I - VIII	Program produces correct output for some input values. Rubric: I - III	Program produces correct output for all positive input values. Rubric: I	Program produces correct output for all positive, negative and zero input values. Rubric: I		10%	